

STA 35C: Statistical Data Science III

Lecture 15: Linear Model Selection – Subset Selection

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Announcements

Homework 4 is due tomorrow (Tue, May 5, 11:59 PM)

- If you have questions, please ask early

Mid-course survey: Thank you for the constructive feedback

- I appreciate your comments/requests on pacing, examples, homework, and exam preparation
 - I will incorporate more worked examples and more guided introductory parts in homework
- Some requests are not feasible, including lecture recordings and weekly review sessions
 - The textbook authors' [lecture slides](#) and [lecture videos](#) may be helpful
- A note on the roles and purposes of homework and exams:
 - Learning statistical data science requires both conceptual understanding and hands-on practice
 - Lecture introduces the key ideas with representative examples
 - Homework builds conceptual depth and computation practice that are more time-consuming
 - Exams test fluency with core concepts, essential computations, and interpretation of results

Agenda

- **Last time: Resampling methods**
 - Cross-validation: *estimate test performance* using training data
 - The bootstrap: *quantify uncertainty* by resampling from the given dataset
- **Model Selection (Today & Wed):** Identify relevant predictors among many
 - **Why?**
 - Improve prediction accuracy (avoid overfitting)
 - Improve model interpretability
 - **How?**
 - Subset selection: search over predictor subsets (today)
 - Regularization: penalize overly complex models (Wed)
 - Dimension reduction: replace predictors by lower-dimensional summaries

Recap: Resampling methods

Given one dataset, we often want to assess model performance or quantify uncertainty

- **Test performance estimation** (e.g., test MSE)
 - We care about performance on new (test) data, but only have a training dataset
 - **Validation set approach:** Hold out part of the data for validation
 - **Cross-validation:** Repeatedly hold out part of the data & aggregate results for a more reliable test performance estimate
- **Uncertainty quantification** (e.g., standard error)
 - We want to gauge variability in parameter estimates
 - If we could draw fresh samples from nature, we'd see how estimates vary
 - **The bootstrap:** Resample with replacement from the observed data (treating it as an empirical proxy) to approximate the sampling distribution of an estimator

(Linear) Model selection

Recall multiple linear regression:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon.$$

- In reality, we might have many predictors, unsure which are truly helpful
- **Example:** **Credit** dataset
 - Response: **balance**
 - Predictors: **income, limit, rating, cards, age, education, own, student, married, region**
- **Goal:** Choose a subset of *relevant* predictors
 - e.g., {**income, limit, age**} or {**rating, education**} or something else

Why model selection?

Two main reasons:

- **Prediction accuracy**

- Overfitting can occur if we use too many predictors
- If $p > n$, we might not even get a unique least squares solution (variance $\rightarrow \infty$)
- Reducing predictors can lower variance and improve generalization

- **Model interpretability**

- Many of the available predictors might not be truly associated with the response
- Including unnecessary predictors can mislead interpretation
- Simpler models are easier to interpret and explain

How to do model selection?

Three key approaches for linear model selection:

- **Subset selection**
 - Search over subsets of predictors to identify a relevant subset
- **Regularization** (to be discussed on Wed)
 - Add a penalty term to least squares formulation that favors “simpler” models
- **Dimension reduction** (not covered in STA 35C)
 - Project the p predictors into a smaller set of $p' \ll p$ linear combinations

Today's focus: **Subset selection**

- Best subset selection
- How to choose the optimal model
- Stepwise selection (greedy approximation)

Best subset selection

Idea: Try *all* subsets of predictors, and pick the one that performs the best

- With p predictors, there are 2^p possible subsets
- Compare models of different sizes carefully (recall R^2 vs. R^2_{adj})

Procedure¹:

- Let $\mathcal{M}_0$ be the null model (no predictors, just intercept)
- For $k = 1, \dots, p$:
 - Fit all $\binom{p}{k} = \frac{p!}{k!(p-k)!}$ models with exactly k predictors
 - Pick the best (lowest RSS or highest R^2) among them, call it $\mathcal{M}_k$
- Finally, select the best among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using a model-selection criterion
 - e.g., adjusted R^2 or cross-validation (more on this later)

¹See [JWHT21, Chapter 6.5.1] for example codes

Best subset selection: Example ($n = 3, p = 2$)

Example

Dataset: 3 points with 2 predictors (X_1, X_2) and a response Y :

$$(X_1, X_2, Y) = (1, 2, 3),$$

$$(X_1, X_2, Y) = (2, 1, 4),$$

$$(X_1, X_2, Y) = (3, 3, 5).$$

Candidate subsets:

- $\mathcal{M}_0$: Null model (intercept only).
- $\mathcal{M}_1^{(X_1)}$: Use X_1 only.
- $\mathcal{M}_1^{(X_2)}$: Use X_2 only.
- $\mathcal{M}_2$: Use (X_1, X_2) .

Best subset selection: Example (Step 1)

Example

Step 1: Fit each model and compute R^2 .

1) $\mathcal{M}_0$: intercept only

- $\hat{\beta}_0 = \bar{Y} = \frac{3+4+5}{3} = 4.$
- $RSS_0 = \sum (Y_i - 4)^2 = 1 + 0 + 1 = 2.$
- $TSS = 2$, and thus, $R_0^2 = 1 - \frac{2}{2} = 0.$

2) $\mathcal{M}_1^{(X_1)}$: one predictor X_1

- $(x_i, y_i) = \{(1, 3), (2, 4), (3, 5)\}; \bar{x} = 2, \bar{y} = 4.$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^3 (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^3 (x_i - \bar{x})^2} = \frac{(-1) \cdot (-1) + 0 \cdot 0 + 1 \cdot 1}{(-1)^2 + (0)^2 + (1)^2} = \frac{2}{2} = 1, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 4 - 1 \cdot 2 = 2.$$

- Thus $\hat{Y} = 2 + 1X_1 \implies$ fitted values $(3, 4, 5).$
- $RSS_1 = 0 \implies R_1^2 = 1.$

Best subset selection: Example (Step 1, cont'd)

Example

...(continued from the previous slide)...

3) $\mathcal{M}_1^{(X_2)}$: one predictor X_2

- $(x_2, y) = \{(2, 3), (1, 4), (3, 5)\}$, $\bar{x}_2 = 2$, $\bar{y} = 4$.

$$\hat{\beta}_1 = \frac{0 + 0 + 1 \cdot 1}{0 + (-1)^2 + 1^2} = \frac{1}{2} = 0.5, \quad \hat{\beta}_0 = 4 - 0.5 \cdot 2 = 3.$$

- $\hat{Y} = 3 + 0.5X_2 \implies \hat{Y} = \{4, 3.5, 4.5\}$.

$$\text{RSS}_2 = (3 - 4)^2 + (4 - 3.5)^2 + (5 - 4.5)^2 = 1 + 0.25 + 0.25 = 1.5, \quad R_2^2 = 1 - \frac{1.5}{2} = 0.25.$$

4) $\mathcal{M}_2$: two predictors (X_1, X_2)

- With 3 points, a model with 2 predictors can fit perfectly if consistent.
- We get $\text{RSS}_{1,2} = 0$, and thus, $R_{1,2}^2 = 1$.

Best subset selection: Example (Step 2)

Example

Step 2: Compare the four candidate models.

Choose the best by adjusted R^2 or a simpler-subset preference.

Recall $R_{\text{adj}}^2 = 1 - \frac{\text{RSS}/(n-k-1)}{\text{TSS}/(n-1)}$:

$$R_{\text{adj}}^2(\mathcal{M}_0) = 0,$$

$$R_{\text{adj}}^2(\mathcal{M}_1^{(X_1)}) = 1,$$

$$R_{\text{adj}}^2(\mathcal{M}_1^{(X_2)}) = -0.5,$$

$\mathcal{M}_2$: undefined due to $n - p - 1 = 0$.

Therefore, we choose $\mathcal{M}_1^{(X_1)}$ (the one with highest adjusted R^2).

Best subset selection: Visualization

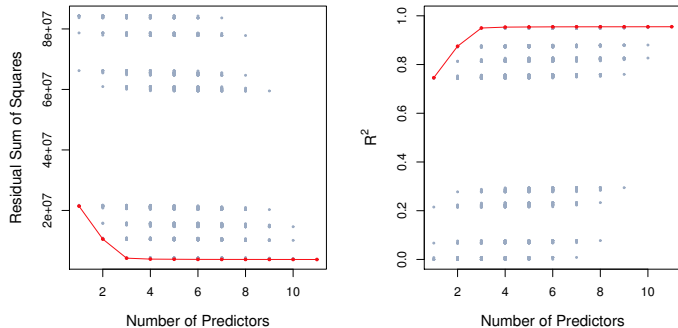


Figure: In the **Credit** dataset, RSS and R^2 are displayed for each subset of the ten predictors. The red frontier indicates the best model at each subset size. The x-axis goes from 1 to 11 because one categorical predictor (three levels) is split into two dummy variables [JWHT21, Figure 6.1].

- Pick the model with the lowest test MSE or best adjusted R^2
- If several models perform similarly (e.g., within 1 SE of the best), pick the simplest

Evaluating models & criteria

Goal: Out of $\{\mathcal{M}_0, \dots, \mathcal{M}_p\}$, choose the model with the best *test* performance

- Training performance (e.g., RSS or R^2) alone can be misleading

Common criteria:

- **Adjusted R^2 :**
 - $R^2_{\text{adj}} = 1 - \frac{\text{RSS}/(n-p-1)}{\text{TSS}/(n-1)}$
 - Increases only if the RSS decrease is large enough to offset the extra predictor
- **Cross-validation:**
 - An empirical approach splitting/re-splitting data to estimate test error
- C_p , **AIC**, **BIC** (beyond the scope of this course):
 - Analytical formulas penalizing model size (p) under certain theoretical assumptions

Visualization of selection criteria

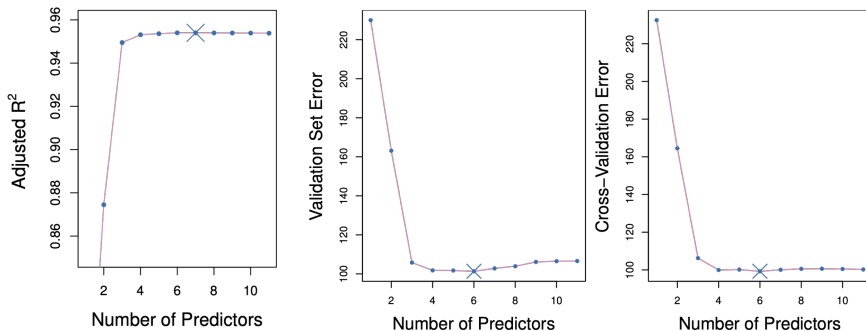


Figure: For the Credit dataset, adjusted R^2 , validation error (single split), and cross-validation error are displayed for the best model containing k predictors, for k ranging from 1 to 11. The overall best model, based on each of these quantities, is shown as a blue cross [JWHT21, Figures 6.2 & 6.3, excerpted].

- These methods often choose similar models

Pop-up quiz: Best subset selection

Question: Which of the following statements about *best subset selection* is correct?

- A) It chooses the model with the smallest training RSS among all subset sizes.
- B) It fits *all* possible models of each size k and picks the best for that k , and also best k .
- C) It can handle extremely large p quickly by skipping some potential subsets.
- D) It always improves test error as more predictors are added.

Answer: B. Best subset selection searches all subsets, keeps the best model of each size, and then uses a criterion beyond training RSS to choose the final size.

Best subset selection: Summary & limitations

Key idea: Exhaustively explore 2^p subsets; pick the best by a test-performance criterion

- Useful when p is small
- $\{\mathcal{M}_k\}$ denotes the best k -predictor model; we choose among $\{\mathcal{M}_0, \dots, \mathcal{M}_p\}$ using a test-performance measure
- Common selection criteria: adjusted R^2 , validation error, cross-validation error, C_p , AIC, BIC
- Straightforward, systematic approach for accuracy & interpretability

Limitation: 2^p grows rapidly (with p), often infeasible for large p

- e.g., $p = 10 \Rightarrow 2^p = 1024$, while $p = 50 \Rightarrow 2^p \approx 10^{15}$ (infeasible)

Forward stepwise selection

Idea: A *greedy*² approximation to best subset selection, adding one predictor at a time

Procedure³:

- $\mathcal{M}_0$: null model with intercept only
- **For** $k = 0, \dots, p - 1$:
 - Consider all $(p - k)$ models that add exactly 1 unused predictors to $\mathcal{M}_k$
 - Pick the best updated model, and call it $\mathcal{M}_{k+1}$
- Finally, compare $\{\mathcal{M}_0, \dots, \mathcal{M}_p\}$ using adjusted R^2 or other test-based metrics

²At each step, pick the best addition via a *local* search

³See [JWHT21, Chapter 6.5.1] for example codes

Forward stepwise selection: Example (Overview)

Example

Dataset: 4 points with 3 predictors (X_1, X_2, X_3) and response Y :

$$(X_1, X_2, X_3, Y) = (1, 2, 2, 2.5),$$

$$(X_1, X_2, X_3, Y) = (2, 1, 1, 3.5),$$

$$(X_1, X_2, X_3, Y) = (3, 3, 2, 6),$$

$$(X_1, X_2, X_3, Y) = (4, 1, 3, 6.5).$$

- Step 0: $\mathcal{M}_0$ fits $Y = \beta_0$; compute $\text{RSS}_0 \approx 11.19$.
- Step 1: Fit $\mathcal{M}_1^{(X_1)}, \mathcal{M}_1^{(X_2)}, \mathcal{M}_1^{(X_3)}$. Pick best single predictor (with largest R^2).
- Step 2: Add a second predictor from the remaining, forming $\mathcal{M}_2$. Check R^2, R_{adj}^2 .
- Step 3: Possibly add the last predictor ($\mathcal{M}_3$ with all three predictors).
- Final selection: Compare $\mathcal{M}_0, \dots, \mathcal{M}_3$ and choose the subset subset with best test performance.

Forward stepwise selection: Example (Step 0)

Example

Step 0: Null model. $\mathcal{M}_0 : Y = \beta_0$.

- $\hat{\beta}_0 = \bar{Y} = \frac{2.5+3.5+6+6.5}{4} = 4.625$.
- $\text{RSS}_0 = \sum (Y_i - 4.625)^2 = (2.5 - 4.625)^2 + (3.5 - 4.625)^2 + (6 - 4.625)^2 + (6.5 - 4.625)^2$
 $= (-2.125)^2 + (-1.125)^2 + (1.375)^2 + (1.875)^2$
 $\approx 4.51 + 1.27 + 1.89 + 3.52$
 $= 11.19$.
- $TSS = 11.19, \quad R_0^2 = 1 - \frac{11.19}{11.19} = 0$.

Step 1: Fit and compare each single-predictor model X_1, X_2, X_3 .

- We now fit $\mathcal{M}_1^{(X_1)}, \mathcal{M}_1^{(X_2)},$ and $\mathcal{M}_1^{(X_3)}$.
- Then compute RSS and R^2 for each; See the next slide for sample calculation for $\mathcal{M}_1^{(X_1)}$.

Conclusion of Step 1: Whichever single predictor yields the highest R^2 (or lowest RSS) is $\mathcal{M}_1$.

Forward stepwise selection: Example (Step 1, further details)

Example

Illustration for X_1 :

- $X_1 = \{1, 2, 3, 4\}$, $Y = \{2.5, 3.5, 6, 6.5\}$.
- Slope $\hat{\beta}_1, \hat{\beta}_0$ are obtained by least squares:

$$\hat{\beta}_1 = \frac{\sum (x_{1i} - \bar{x}_1)(y_i - \bar{y})}{\sum (x_{1i} - \bar{x}_1)^2} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{x}_1 = 2.5$ and $\bar{y} = 4.625$. Eventually, we find

$$\hat{\beta}_1 \approx 1.2, \quad \hat{\beta}_0 \approx 3.025 \text{ (approx).}$$

- Then $\text{RSS}_1 \approx 2.55$, $R_1^2 = 1 - \frac{2.55}{11.19} \approx 0.77$.

Similarly for X_2, X_3 : $\text{RSS}_2 \approx 4.12$, $R_2^2 \approx 1 - \frac{4.12}{11.19} = 0.63$, and $\text{RSS}_3 \approx 3.20$, $R_3^2 \approx 1 - \frac{3.20}{11.19} = 0.71$.

Pick the best single predictor = X_1 , which yields highest R^2 or lowest RSS.

Forward stepwise selection: Example (Step 2)

Example

Step 2: Add a second predictor to $\mathcal{M}_1$

Now $k = 1$. Our model has X_1 , so the unused are X_2 and X_3 :

$$\mathcal{M}_2^{(X_1, X_2)}, \quad \mathcal{M}_2^{(X_1, X_3)}.$$

We fit each, compute RSS & R^2 . For instance:

(1) $\mathcal{M}_2^{(X_1, X_2)}$: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$. After fitting this model:

$$\text{RSS}_{1,2} \approx 1.80, \quad R_{1,2}^2 = 1 - \frac{1.80}{11.19} \approx 0.84.$$

(2) $\mathcal{M}_2^{(X_1, X_3)}$: Similarly, we get $\text{RSS}_{1,3} \approx 1.40$, $R_{1,3}^2 = 1 - \frac{1.40}{11.19} \approx 0.875$.

Hence $\mathcal{M}_2 = \mathcal{M}_2^{(X_1, X_3)}$ (larger R^2).

Forward stepwise selection: Example (Step 3 & Selection)

Example

Step 3: Add the remaining predictor to $\mathcal{M}_2$

Now $k = 2$. Our model includes X_1 and X_3 . The remaining predictor is X_2 . So we consider:

$$\mathcal{M}_3 : Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3.$$

- Fit the full model with 3 predictors, compute $\text{RSS}_{1,2,3}$, $R_{1,2,3}^2$
- Suppose $\text{RSS}_{1,2,3} \approx 1.25$, $R_{1,2,3}^2 = 1 - \frac{1.25}{11.19} \approx 0.89$.

Finally, we might pick $\mathcal{M}_1$ or do cross-validation among the four models:

$\mathcal{M}_0 :$	$R^2 = 0,$	$R_{\text{adj}}^2 = 0,$
$\mathcal{M}_1 = \{X_1\} :$	$R^2 \approx 0.77,$	$R_{\text{adj}}^2 \approx 0.66,$
$\mathcal{M}_2 = \{X_1, X_3\} :$	$R^2 \approx 0.875,$	$R_{\text{adj}}^2 \approx 0.63,$
$\mathcal{M}_3 = \{X_1, X_2, X_3\} :$	$R^2 \approx 0.89,$	$R_{\text{adj}}^2 \text{ undefined } (n - p - 1 = 0).$

Pop-up quiz: Best subset vs. forward stepwise

Question: In *forward stepwise selection*, how do we choose a new predictor at each step?

- A) By removing whichever predictor contributes least to the model's fit.
- B) By guessing randomly among the remaining predictors.
- C) By trying each unused predictor one at a time and selecting the one that yields the best improvement in the chosen metric (e.g., R^2).
- D) By fitting all 2^p possible models and picking the global best.

Answer: C. Forward stepwise selection tries each *remaining* predictor individually at each step, then adds the one that most improves the model.

Example: Stepwise selection may yield a different subset

# Variables	Best subset	Forward stepwise
One	rating	rating
Two	rating, income	rating, income
Three	rating, income, student	rating, income, student
Four	cards, income student, limit	rating, income, student, limit

Figure: The first four chosen models for best subset selection and forward stepwise selection on the **Credit** dataset. The first three models are identical, but the fourth differs [JWHT21, Table 6.1].

- Stepwise typically performs well and is computationally much cheaper:

$$1 + \sum_{k=0}^{p-1} (p - k) = 1 + \frac{p(p+1)}{2} \ll 2^p$$

- However, it may pick a different subset if the greedy path diverges

Illustration: Best subset selection vs. forward stepwise

An illustrative example with $p = 4$:

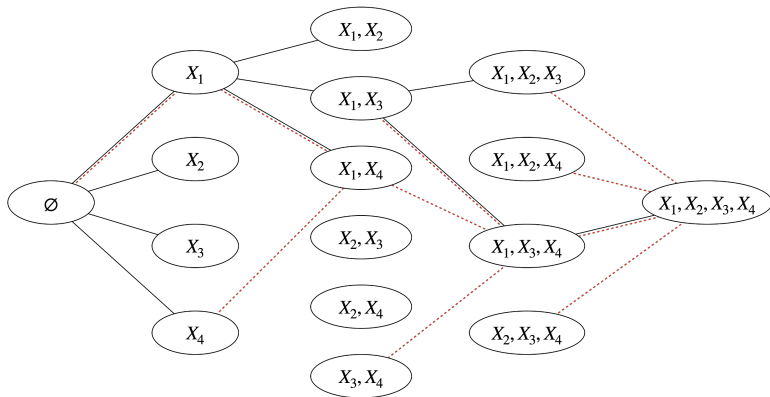


Figure: Illustration of forward stepwise (**black solid path**; successively adding $X_1 \rightarrow X_4 \rightarrow X_3 \rightarrow X_2$) and backward stepwise (**red dashed path**; removing $X_2 \rightarrow X_3 \rightarrow X_4 \rightarrow X_1$). Best subset selection checks all 2^p possibilities; all three can yield different outcomes.

Backward stepwise selection

Backward stepwise selection is similar to the forward selection, but reverses the direction of the search

Backward stepwise selection:

- Start with the full model; remove one predictor at a time
- Usually require $n > p$ so the full model can be fit initially

Comparison with forward stepwise:

- Both are *greedy* algorithms using local decisions
- Both drastically reduce the search space vs. best subset when p is large
- They can yield different subsets if they take different paths

Wrap-up & Takeaways

Model selection (Subset selection): Identify a subset of relevant predictors

- **Purposes:**

- Improve prediction accuracy and avoid overfitting
- Enhance model interpretability

- **Methods:**

- **Best subset selection:**

- Exhaustively checks all 2^p subsets (optimal but expensive)
- Feasible only for small p (e.g., $p \lesssim 20$)

- **Stepwise selection** (forward or backward):

- Much fewer model fits needed: $1 + \frac{p(p+1)}{2}$ vs. 2^p
- Often performs well in practice, but may miss the globally optimal subset

- **Overall:**

- Stepwise methods generally give good models but are *not guaranteed* to be optimal
- For moderate or large p ($\gtrsim 50$), stepwise is typically the only feasible approach

References



Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.

An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.

Springer, New York, NY, 2nd edition, 2021.